

Low molecular weight marker

Amount	Ingredient	Stock	Final
1.25 mL	Sample buffer	4 ×	1 ×
1 mg	Lysozyme, hen egg white [9001-63-2]	14 kDa	0.2 g/L
1 mg	Trypsin inhibitor, soybean [9035-81-8]	22 kDa	0.2 g/L
1 mg	Carbonic anhydrase, from bovine erythrocytes [9001-03-0]	29 kDa	0.2 g/L
2 mg	Peroxidase, horseradish [9003-99-0]	40 kDa	0.4 g/L
1 mg	Albumin, bovine serum [9048-46-8]	66 kDa	0.2 g/L
2 mg	Lipoxidase, soybean [9029-60-1]	96 kDa	0.4 g/L
To 5 mL	Water, reagent-grade		

Dilute each component in 500 μ L 1 × sample buffer. Dispense into 20 μ L aliquots. Store at -20°C . **Note:** Check the integrity of each component on a gel before combining. Apply 20 μ L low molecular weight marker (LMW) per lane.

Low molecular weight marker

1 × Sample buffer, 0.2 g/L Lysozyme, 0.2 g/L Trypsin inhibitor, 0.2 g/L Carbonic anhydrase, 0.4 g/L Peroxidase, 0.2 g/L Albumin, 0.4 g/L Lipoxidase

Date: Sign: R0110

High molecular weight marker

Amount	Ingredient	Stock	Final
1.25 mL	Sample buffer	4 ×	1 ×
2 mg	Peroxidase, horseradish [9003-99-0]	40 kDa	0.4 g/L
1 mg	Albumin, bovine serum [9048-46-8]	66 kDa	0.2 g/L
2 mg	Lipoxidase, soybean [9029-60-1]	96 kDa	0.4 g/L
2 mg	β -Galactosidase, from <i>E. coli</i> [9031-11-2]	116 kDa	0.4 g/L
2 mg	Myosin, heavy chain, from rabbit muscle [1638095-42-7]	205 kDa	0.4 g/L
To 5 mL	Water, reagent-grade		

Dilute each component in 500 μ L 1 × sample buffer. Dispense into 20 μ L aliquots. Store at -20°C . **Note:** Check the integrity of each component on a gel before combining. Apply 20 μ L high molecular weight marker (HMW) per lane.

High molecular weight marker

1 × Sample buffer, 0.4 g/L Peroxidase, 0.2 g/L Albumin, 0.4 g/L Lipoxidase, 0.4 g/L β -Galactosidase, 0.4 g/L Myosin

Date: Sign: R0111

Resources

Casting SDS polyacrylamide gels

Gel type		Tris-Glycine							Tris-Acetate*		Bis-Tris*						
Gel concentration		4-20%	8-16%	10-20%	8%	10%	12%	15%	3-8%	7%	4-12%	10%		12%			
Running buffer		Tris-Glycine							Tris-Acetate		MOPS	MES	MOPS	MES	MOPS	MES	
		Apparent Molecular Weights, kDa															
% length of gel	10																
	20	250	250	250	250	250	250	250	205	205	185	190	185	190	185	190	
	30	130	130	130	130	130	130	130			115	115	115	115	115	115	
	40	100	100	70	100	100	100	100			80	80	80	80	80	80	
	50	70	70	55	70	55	35	25			65	50	50	50	50	30	
	60	55	55	35	55	35	25				50	30	30	25	30	25	
	70	35		25		25					30	25		25			
	80	25	35		35		15	15			30		25		25	15	
	90	15	25	15	25						25	15		15			
	100	10	10	10	15	15	10	10	25	25	10	10	15	10	15	10	

In Step 1: Migration and resolution of denatured proteins in different buffer systems and varying acrylamide concentrations established by a commercial protein ladder. Prestained proteins can exhibit different mobility than the unstained standard. Reproduced from Thermo Scientific Pub. No. MAN0011773.

List of references

Change log

- 2022-09-20 Benjamin C. Buchmuller Adaptation as SOP; added SDS/Laemmli and LDS sample buffer recipes.
- 2023-01-13 Benjamin C. Buchmuller Added low and high molecular weight marker recipes from C. R. Hauck lab.
- 2024-03-11 Benjamin C. Buchmuller Corrected bromophenol blue content in loading dye concentrate to 0.02% (w/v); literature ranges 0.01–0.50%; less is better.
- 2024-04-21 Benjamin C. Buchmuller Corrected diluted molar concentrations of MES and MOPS running buffers to 50 mM; added a section on crosslinker ratios.
- 2024-07-17 Benjamin C. Buchmuller Adjusted volumes of resolving gels to accommodate three gels with 1.5 mm spacer or four gels with 1.0 mm.

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